



UNIT 1: SOFTWARE — SIMPLE STUDY NOTES

SECTION 1: WHAT IS SOFTWARE?

1. The Basics of Software

- **Definition:** Software is the collection of programs that run on a computer.
- **How it works:** Programs are lists of instructions that the computer's processor carries out (executes). The processor then tells the rest of the computer what to do.
- **Running programs:** Some computers can only run one program at a time, while others can run many programs at the same time.

2. Types of Computers by Purpose

- **Single-Purpose Computers:** These can only run one specific piece of software. For example, the computer inside a microwave only runs the program that controls the microwave.
- **General-Purpose Computers:** These can do many different tasks because you can install different software on them. Examples include tablets, smartphones, and laptops.

3. The Two Main Groups of Software

All software fits into one of these two basic groups:

- **System Software:** Programs made to control, run, and take care of the computer hardware.
- **Application Software:** Programs that help users do normal tasks or make things, like writing an essay or editing a photo.

SECTION 2: UTILITY SOFTWARE

Utility software is a type of system software. Utilities do everyday maintenance and configuration tasks to keep the computer running well.

1. Backup Utilities

- **What it does:** Makes a copy of your files and programs so you do not lose them if the computer breaks.
- **How it works:** You can start a backup yourself, or set it to run automatically. Automatic backups usually run when no one is using the computer (like at night).

2. Disk Defragmentation (Cleaning up Hard Drives)

- **The Problem (Fragmentation):** When you save files to a magnetic hard disk drive (HDD), the computer puts the data into any empty space it can find. Over time, files get

broken into pieces (fragmented) and scattered out of order across the disk. When you try to open the file, the drive's mechanical read/write head has to search all over the disk to find the pieces, which slows down the computer.

- **The Solution:** A defragmentation utility reorders the pieces and puts them back in order, close together. This means the drive head does not have to search for pieces, which speeds up how fast files load. This usually runs automatically in the background.

3. Compression Utilities (Shrinking Files)

- **What it does:** Reduces the size of a file so it takes up less storage space.
- **Run-Length Encoding (RLE):** A simple way to compress files by looking for repeating patterns. Instead of saving every single repeated item, it saves the item once and counts how many times it repeats.
 - *Example:* A long file that looks like aaaaaaaaaabbbbbbbcc (16 letters) is compressed down to a8b6c2 (6 characters), using much less space.
- **Decompression:** Compressed files cannot be opened by normal apps. You must use the utility to **decompress** (extract) the file back to its original size before using it again.

4. Disk Formatting Utilities

- **What it does:** Prepares a new storage device, like a hard drive or USB flash drive, so it can be used for the first time.
- **Warning:** If you format a drive that already has files on it, all the data will become unreadable by normal apps. Formatting utilities will always ask you to confirm that you are sure before they start.
- **Erasing Data:** Quick formatting does not erase data completely; people can still find it using special recovery tools. To wipe it completely, you need a utility that overwrites the data with new code.

SECTION 3: OPERATING SYSTEMS (OS)

The Operating System is the main program that lets you control and manage the computer's hardware.

Human Users ↔ Application Software ↔ Operating System ↔
Hardware [1]

1. Types of Operating Systems

- **Single-User OS:** Made for one person to use at a time. It does not let different people change how the screen looks or keep private profiles. These are usually found in household appliances.
- **Network OS:** Used in schools and businesses where many computers connect together. It can:
 - Check your username and password when you log in.
 - Keep accounts separate so users cannot see or touch each other's files.
 - Give users access to shared network storage and shared printers.

2. Main Tasks of the OS

A. Memory Management

The OS gives out temporary memory (RAM) to apps when they need it. When you close the app, the OS takes that memory back so other apps can use it.

B. Resource Management

Resources include internal parts like the processor (CPU) and graphics card, plus outside devices like printers. The OS makes sure apps get to use these resources when needed. If an app tries to use a resource that is already busy, the OS puts the request into a waiting line (queue).

C. Security

The OS protects your computer by using **authentication** (checking passwords, fingerprints, or access cards) before letting people use the system. It also provides a **firewall** to block unauthorized people from accessing your computer over a network.

D. Print Spooling

Computers send pages to a printer much faster than a printer can actually print them. To stop the computer from freezing up, the OS uses **print spooling** to keep the pages waiting in an orderly queue until the printer is ready for them.

SECTION 4: APPLICATION SOFTWARE

Application software (apps) includes the programs that let you do specific tasks, like creating a presentation, typing a letter, or playing a game.

 **EXAM TIP:** In the exam, **never** write brand names like *Microsoft Word* or *Excel*. Always use the generic type name, like **word processing software** or **spreadsheet software**.

1. Office Productivity Tools

These are often sold together as an **application suite** (a group of apps that look similar and make it easy to move data between them).

- **Word Processing Software:** Used to create documents that are mostly text, like letters, essays, and reports. They let you change fonts (text size and style), check your spelling automatically, leave comments, and track changes when working with other people.
- **Desktop Publishing (DTP) Software:** Similar to word processors, but much better at handling complicated page layouts. It helps designers see exactly what a magazine or brochure layout will look like when printed.
- **Spreadsheet Software:** Used to do math and automatic calculations. You use **formulas and functions** to calculate numbers. They are great for managing budgets and answering "what if?" questions (like, "What happens to my profits if I change the price of

tickets?").

- **Database Management Systems (DBMS):** Used to enter, edit, and search through large amounts of data. They use **forms** to type in data easily, **queries** to search for specific files, and **report builders** to print out clean summaries.

2. Creative and Specialized Software

- **Web Authoring Software:** Used to build web pages by creating **HTML** code. Web browsers (like Chrome or Firefox) read HTML and turn it into visual web pages. Many tools use a **WYSIWYG** (*What You See Is What You Get*) editor, which lets you design visually without writing code.
- **Graphics Editors (Vector Graphics):** Used for drawings and logos made out of mathematical lines and shapes. They use layers and let you resize shapes without making the image blurry.
- **Photo Editors (Bitmap Images):** Used to edit digital photographs, which are made out of tiny dots called pixels. Features include fixing brightness, blurring backgrounds, and **cropping** (cutting off the outer edges of a picture).
- **Sound Editing Software:** Used to mix audio files, cut clips together, or add music soundtracks to video. Features include volume control, changing the **tempo** (speed of the sound), **muting** tracks, and **normalization** (making sure the volume is set to a good level).
- **Presentation Software:** Used to make visual slideshows to support a speech. You can add text, images, and videos onto pages called **slides**.
- **Control Applications:** Software that takes data from sensors (like temperature or movement sensors) and uses it to control physical devices called actuators (like motors). For example, control software automatically opens elevator doors when it senses the lift has stopped.
- **Project Management Software:** Helps managers plan out project tasks on a timeline. It uses **Gantt charts** to show which tasks depend on other tasks, track budgets, and highlight **milestones** (important target dates).

3. Communication Software

Used to connect with people over the internet and share text, images, sound, and video.

- **Web Browsers:** Let you view web pages and exchange files using **FTP** (File Transfer Protocol).
- **Email:** Can be accessed via local apps on your computer or webmail through a browser. They use labels and filters to sort your mail automatically.
- **SMS, MMS, and Instant Messaging:**
 - **SMS:** Standard text messages sent over phone networks. Limited to 160 characters; no internet needed.
 - **MMS:** Like SMS, but can send longer text, photos, and videos without internet.
 - **Instant Messaging:** Apps that let you chat over the internet. They show real-time features like typing indicators and location sharing.

SECTION 5: LICENSING AND UPDATES

1. Software Licenses

You need a legal **software license** to install and use software. Using software without a license is illegal and called software piracy. There are two main types of software:

- **Free or Open-Source Software:**
 - *Free software* means users have the complete **freedom** to change, copy, or share the program without restrictions (it does not always mean it costs zero money).
 - *Open-source* means the developer shares the raw **source code** so anyone can look inside and modify how the program works.
- **Proprietary Software:** Software owned by a specific brand or company. The source code is hidden, and you cannot copy or modify it. When proprietary software is given away for free, it is called **freeware**.

2. Software Updates

Developers release updates over the internet for several reasons:

1. To fix security issues and patch code **bugs** (programming errors).
2. To make the app work better with new operating systems.
3. To add cool new features and make the app easier to use.

 **SAFETY RULE:** Installing updates can sometimes cause unexpected problems or errors. **Always run a full backup of your files before you update your software.**